

Compiler engineer with professional LLVM and C# static-analysis experience. At MCST, I worked on LLVM 22 loop/interprocedural analysis, legality, and IR transformations; at ISP RAS, I contribute to SharpChecker, an industrial C#.NET static-analysis platform. Independent work spans conservative LLVM transformations, Roslyn fixed-point data flow, and UniversalToolchain, a compiler/language-infrastructure project built around typed artifact contracts and deterministic planning; an architecture talk based on the project was accepted to LangDev'26.

Professional compiler / analysis
MCST: LLVM passes, loop/interprocedural analysis, and legality; ISP RAS: SharpChecker C#.NET static analysis.

Compiler architecture
UniversalToolchain: typed artifacts; dependencies/conflicts, provider selection, pass order, artifact routes, and backends compile into an immutable LanguagePlan before execution.

Verification & backends
Conservative rejection paths, LLVM Verifier and before/after execution; interpreter/CIL parity; a separate x86-64 codegen/register-allocation pipeline.

EXPERIENCE

- 2026 — present

ISP RAS — Static Analysis Engineer
 - Contribute to SharpChecker, ISP RAS's industrial static-analysis platform for C#.NET.
- July — August 2026

MCST — Compiler Engineering Intern
 - Implemented an LLVM LICM pass with loop analysis, side-effect/speculative-safety checks, and hoisting of loop-invariant instructions to preheaders.
 - Developed a conservative interprocedural global-IV prototype with APInt affine evolution, transitive call effects, a separate legality phase, and LLVM IR transformation; unsupported CFG/call cases are rejected.

PROJECTS

- LLVM Interprocedural Global IV Optimization**

A conservative LLVM 22 prototype/MVP with an actual IR transform; 29 positive/negative regression cases cover verifier validity, idempotence, and before/after execution.
- UniversalToolchain**

Current exact test manifest: 1,324/1,324. In a frozen-corpus verifier experiment, B2 detected 32/32 primary and 10/10 challenge operators with 0/100 false positives on valid controls; four post-freeze holdouts were checked separately.
- DerefAfterNullAnalyzer**

Diagnostics for potentially unsafe dereference paths over the real CFG, with successor reprocessing until the data-flow state reaches a fixed point.
- x86-64 Codegen & Register Allocation Lab**

A reference interpreter and differential execution check generated-code semantics; the allocator assignment contract is verified independently.

SKILLS

- Compilers**

C++, LLVM 22, LLVM IR, optimization passes, interprocedural analysis, CFG/SSA, dominators, loop analysis, LICM, legality analysis
- Static / Program Analysis**

C#, .NET, Roslyn ControlFlowGraph, fixed-point data-flow analysis, state propagation and joins
- Compiler Infrastructure**

typed artifact contracts, IR/lowering pipelines, deterministic planning/pass ordering, backend/runtime composition, executable verification
- Codegen & Systems**

Rust, SysV x86-64, liveness/interference, register allocation, Linux, CMake

EDUCATION

HSE University — Software Engineering, 2026–2030

RECOGNITION

- Accepted talk on UniversalToolchain's architecture; LangDev'26 takes place October 8–9, 2026 in Málaga.
- Moscow / remote